Servlets
Sample Programs

NOTE

- All the Servlets(Java classes) should be created inside src/main/java folder.
- All html , css files should be created inside Webapp folder.

1 Creating Servlet By Extending GenericServlet Class.

First.html

```
<body>

<form action="demo">

NAME : <input type="text" name="myname">

<input type="submit">

</form>

</body>
```

FirstServlet.java

```java
// 3. specify the URL for accessing this servlet
@WebServlet("/demo")
//1. create a class by extending GenericServlet class
public class FirstServlet extends GenericServlet
{
    // 2. override service()
    @Override
    public void service(ServletRequest req, ServletResponse res)
                                throws ServletException, IOException
    {

        // fetch the data sent by the client
        String name =   req.getParameter("myname");

        // creating PrintWriter object for generating response of this servlet
        PrintWriter pw = res.getWriter();

        pw.print("<h1>WELCOME TO MY FIRST SERVLET PROGRAM..."+name+"</h1>");

    }
}
```

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Details.html

```
<body>
<h2>PERSONAL DETAILS FORM</h2>

<form action="mydetails">

NAME : <input type="text" name="myname"><br><br>
AGE : <input type="number" name="myage"><br><br>
PHONE NUMBER : <input type="number" name="myphone"><br><br>
ADDRESS : <input type="text" name="myaddress"><br><br>
DATE OF BIRTH : <input type="date" name="mydate"><br><br>

<input type="submit">

</form>
</body>
```

DetailsServlet.java

```java
@WebServlet("/mydetails")
public class DetailsServlet extends GenericServlet
{
    @Override
    public void service(ServletRequest req, ServletResponse res)
                        throws ServletException, IOException
    {

        String name = req.getParameter("myname");
        String age = req.getParameter("myage");
        String phone  = req.getParameter("myphone");
        String address = req.getParameter("myaddress");
        String dob = req.getParameter("mydate");

        PrintWriter pw = res.getWriter();
        pw.print("<html><body>");
        pw.print("<h1>YOUR DETAILS</h1>");
        pw.print("<br><br><b>NAME : </b>"+name);
        pw.print("<br><br><b>AGE : </b>"+age);
        pw.print("<br><br><b>ADDRESS : </b>"+address);
        pw.print("<br><br><b>PHONE NUMBER : </b>"+phone);
        pw.print("<br><br><b>DATE OF BIRTH : </b>"+dob);
        pw.print("</body></html>");
    }

}
```

3 Use of RequestDispatcher interface.

Login.html

```
<body>
<h1>LOGIN HERE...</h1>

<form action="login">

USERNAME : <input type="text" name="uname"><br><br>
PASSWORD : <input type="text" name="password"><br><br>
<input type="submit" value="LOGIN">

</form>

</body>
```

LoginServlet.java

```java
@WebServlet("/login")
public class LoginServlet extends GenericServlet {

    @Override
    public void service(ServletRequest req, ServletResponse res)
                                    throws ServletException, IOException
    {
        String username = req.getParameter("uname");
        String password = req.getParameter("password");

        // validation --> username = suresh and password = 12345
        if(username.equals("suresh") && password.equals("12345"))
        {
            RequestDispatcher rd = req.getRequestDispatcher("display");
            rd.forward(req, res);
        }
        else
        {
            res.setContentType("text/html");

            PrintWriter pw = res.getWriter();
            pw.print("INCORRECT USERNAME OR PASSWORD...!");
            RequestDispatcher rd = req.getRequestDispatcher("login.html");
            rd.include(req, res);
        }
    }


}
```

DisplayServlet.java

```java
@WebServlet("/display")
public class DisplayServlet extends GenericServlet{

    @Override
    public void service(ServletRequest req, ServletResponse res)
                                throws ServletException, IOException
    {

        String username = req.getParameter("uname");
        PrintWriter pw = res.getWriter();
        pw.print("<h1>WELCOME ..."+username+"</h1>");
    }
}
```

4 Creating Servlet By Extending HttpServlet Class.

- **Get request is sent by the client which is handled by servlet using doGet() method.**

sample.html

```
<body>
<form action="sample">

NAME : <input type="text" name="myname"><br><br>
<input type="submit">

</form>
```

SampleServlet.java

```java
@WebServlet("/sample")
public class SampleServlet extends HttpServlet
{
    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp)
                                                    throws ServletException, IOException
    {

            String name = req.getParameter("myname");
            PrintWriter pw = resp.getWriter();
            pw.print("Welcome..."+name);
    }
}
```

5 Creating Servlet By Extending HttpServlet Class.

- **Post request is sent by the client which is handled by servlet using doPost() method.**

test.html

```
<body>
<form action="test" method="post">

NAME : <input type="text" name="myname"><br><br>
AGE : <input type="number" name="myage"><br><br>
PHONE NUMBER : <input type="number" name="myphone"><br><br>
ADDRESS : <input type="text" name="myaddress"><br><br>
DATE OF BIRTH : <input type="date" name="mydate"><br><br>
<input type="submit">

</form>
</body>
</html>
```

TestServlet.java

```java
@WebServlet("/test")
public class TestServlet extends HttpServlet
{
    @Override
    protected void doPost(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOException {

        String name = req.getParameter("myname");
        String age = req.getParameter("myage");
        String phone  = req.getParameter("myphone");
        String address = req.getParameter("myaddress");
        String dob = req.getParameter("mydate");

        PrintWriter pw = resp.getWriter();
        pw.print("<html><body>");
        pw.print("<h1>YOUR DETAILS</h1>");
        pw.print("<br><br><b>NAME : </b>"+name);
        pw.print("<br><br><b>AGE : </b>"+age);
        pw.print("<br><br><b>ADDRESS : </b>"+address);
        pw.print("<br><br><b>PHONE NUMBER : </b>"+phone);
        pw.print("<br><br><b>DATE OF BIRTH : </b>"+dob);
        pw.print("</body></html>");

    }
}
```

6 Use of sendRedirect() method.

search.html

```
<body>

<form action="search">

Search : <input type="text" name="name">
<input type="submit">

</form>
</body>
</html>
```

SearchServlet.java

```java
@WebServlet("/search")
public class SearchServlet extends HttpServlet
{
    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOException {

        String data = req.getParameter("name");
        if(data.equalsIgnoreCase("java"))
        {
            // redirecting response to java tutorial of javatpoint
            resp.sendRedirect("https://www.javatpoint.com/java-tutorial");
        }
        else
        {
            // redirecting response to google home page
            resp.sendRedirect("https://www.google.com/");
        }
    }
}
```

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Example of cookies

demo.html

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>

<form action="demo">

NAME : <input type="text" name="myname"><br><br>
<input type="submit">

</form>

</body>
</html>
```

DemoServlet.java

```java
@WebServlet("/demo")
public class DemoServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOException {

        String name = req.getParameter("myname");

        // creating a cookie
        Cookie ck = new Cookie("username", name);

        // adding cookie to response
        resp.addCookie(ck);

        PrintWriter pw = resp.getWriter();
        pw.print("<html><body>");
        pw.print("<h2>Cookie has been created successfully...</h2><br><br>");
        pw.print("<a href='read'>CLICK HERE TO READ THE COOKIE</a>");
        pw.print("</body></html>");
    }
}
```

ReadServlet.java

```java
@WebServlet("/read")
public class ReadServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOException {

        PrintWriter pw = resp.getWriter();

        // getting the cookie which is sent along with request.
        Cookie [] cookies = req.getCookies();

        // checking if cookie exists or not
        if(cookies != null) {
            pw.print("Cookie found...");

            for(Cookie ck : cookies) {
                pw.print("NAME : "+ck.getName()+" VALUE : "+ck.getValue());
            }
        }
        else {
            pw.print("No cookie found..");
        }
    }

}
```

Example of HttpSession

demo.html

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>

<form action="demo">

    NAME : <input type="text" name="myname"><br><br>
    AGE : <input type="number" name="myage"><br><br>

    <input type="submit">

</form>

</body>
</html>
```

```java
@WebServlet("/demo")
public class DemoServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOException {

        String name = req.getParameter("myname");
        String age = req.getParameter("myage");

        HttpSession session = req.getSession();
        session.setAttribute("username", name);

        PrintWriter pw = resp.getWriter();
        pw.print("<html><body>");
        pw.print("<h2>HttpSession has been created successfully.</h2><br><br>");
        pw.print("<a href='read'>CLICK HERE TO READ SESSION</a>");
        pw.print("</body></htm>");
    }
}
```

```java
@WebServlet("/read")
public class ReadServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOException {

        PrintWriter pw = resp.getWriter();

        // getSession(false) returns Returns the current HttpSession associated with this request or,
        // if there is no current session , it return null.
        HttpSession session = req.getSession(false);

        if(session != null) {

            pw.print("Request coming from same client");
            // session.invalidate(); invalidates the current session and removes data linked to this session. (used for logout)
        }
        else {
            pw.print("Request has come from a different client");
        }
    }

}
```